

hw0

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1. (a) after renormalizing $p_X(x) = \sum_y p_{X,Y}(x) = 1/3$ for all x . similarly,
 $p_Y(y) = \sum_x p_{X,Y}(x) = 1/2$ for all y
 (b) $\frac{P_{X,Y}(2,1)}{P_Y(1)} = 1/3$
 (c) yes. each probability is constant.
 (d)

$$\mathbb{E}(x) = \sum_x x * 1/3 = 1 \quad (1)$$

$$\mathbb{E}(y) = \sum_y y * 1/2 = 0.5 \quad (2)$$

2. (a)

$$\begin{aligned} P(Y = 1) &= P(Y = 1|X = 0)P(x = 0) + P(Y = 1|X = 1)P(x = 1) \\ &= \alpha(1 - p) + \beta p \\ P(Y = 0) &= 1 - \alpha(1 - p) + \beta p \end{aligned} \quad (3)$$

- (b)

$$\begin{aligned} P(X = 1|Y = 1) &= P(X = 1)P(Y = 1|X = 1)/P(Y = 1) \\ &= \beta p / (\alpha(1 - p) + \beta p) \end{aligned} \quad (4)$$

- (c) $\alpha = \beta$

- (d)

3. (a)

$$P(Y = 1) = P(Y = 0) = 0.5 \quad (5)$$

- (b) if $Y=0$, then $x=1,2 = 1/2$. $x = 3,4 = 0$. if $Y=1$, then $x=3,4 = 1/2$.
 $x = 1,2 = 0$.
 (c) y is surjective but not injective.
 (d) no. full invertability would be required

- 4.